MCO-1D ORNITHOPTER

Commissioning Manual — Revision 14

This is the complete commissioning manual for the MCO-1D eight-wing metachronal ornithopter. It covers every step from first assembly to experimental discovery flights, written assuming no prior knowledge of the project. It reflects the corrected architecture (8 independent motor-sector assemblies), full-state per-wing sensing, and all optimizer improvements from that sensing.

Aircraft	MCO-1D metachronal cascade ornithopter
MTOW	3.448 kg (7.60 lb)
Wings	8 independent (4 tandem pairs W1-W4)
Motors	8 × GoolRC 5010-14 360KV (one per wing)
ODrives	4 × v3.6 (axis0=port, axis1=stbd per pair)
Stall speed	4.85 m/s (9.4 knots) out-of-ground-effect
VNE	28.0 m/s (54.4 knots)
Freq. ceiling	3.0 Hz (hard limit in software)
Per-wing sensing	AoA, Sweep (MCP3208), Stroke (ODrive)
Wave Library	44,127 lines Python
Hypotheses	24 active, 90 variants
Tick rate	50 Hz (20ms per tick)
Revision	14 — July 2026

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PART I — ORIENTATION

SECTION 1 — WHAT THIS AIRCRAFT IS AND WHY IT WAS BUILT

1.1 The Aircraft

The MCO-1D is a prototype ornithopter — an aircraft that flies by flapping its wings. It has eight wings in four tandem pairs arranged front to rear. Each wing is completely independent: its own motor, its own sector wheel, its own pitch servo controlling angle of attack, and its own sweep servo controlling fore-aft position. No wing shares any mechanical drive with any other wing. The aircraft weighs 3.448 kilograms and cannot stall in the conventional fixed-wing sense because it generates lift by active flapping rather than airspeed over a fixed surface.

1.2 Why This Aircraft Exists

The MCO-1D is a testbed for a Search and Rescue aircraft lineage. SAR missions require hovering over narrow ledges, descending into confined spaces, and operating in wind that grounds conventional drones. A flapping-wing aircraft with eight independent wings has qualitatively different capabilities: it can reconfigure in real time, and it has eight separate sources of lift and control authority where a quadrotor has only four.

1.3 What Makes This Design Unusual

Every wing on the MCO-1D has five independently controllable parameters: amplitude, frequency, angle of attack (AoA), sweep, and dihedral. With eight wings this gives forty independent parameters changeable every 20 milliseconds. The flight software — the Wave Library — includes an optimizer that continuously discovers which parameter combinations work best for each flight condition. It also contains a library of 24 biological hypotheses drawn from dragonflies, plesiosaurs, sea turtles, bats, and sea lions, tested automatically during dedicated discovery flights.

SECTION 2 — THE ARCHITECTURE: 8 INDEPENDENT WING ASSEMBLIES

■ WARNING: ARCHITECTURE CORRECTION from earlier revisions: Each wing has its own independent motor and sector wheel. There is NO shared rope drive between port and starboard wings on any pair. This is 8 independent motor-sector assemblies, not 4 shared assemblies.

2.1 Wing Numbering

W1 port / W1 starboard	Frontmost pair. Leads the metachronal cascade. ODrive board 0: axis0=W1p, axis1=W1s.
W2 port / W2 starboard	Second pair. Primary lift generator in cruise. ODrive board 1: axis0=W2p, axis1=W2s.
W3 port / W3 starboard	Third pair. Wake capture from W2. ODrive board 2: axis0=W3p, axis1=W3s.
W4 port / W4 starboard	Rear pair. Cascade closure, endurance role. ODrive board 3: axis0=W4p, axis1=W4s.

2.2 The Metachronal Cascade

The four pairs do not flap simultaneously. W1 leads, then W2 follows after a phase offset, then W3, then W4. This staggered timing allows each rearward pair to capture energy from the wake shed by the pair ahead — the same mechanism studied in plesiosaurs, where hind flippers generated 60% more thrust when correctly phased with the forward flippers than when operating alone.

2.3 Per-Wing Independence

Because every wing has its own motor, each can be commanded to any frequency independently. Port and starboard on the same pair can run at different frequencies, different amplitudes, and different AoA simultaneously. This is the capability that enables the novel hypotheses H-24, H-25, and H-26 that have no biological precedent and no equivalent in any conventional aircraft.

SECTION 3 — HOW THE SOFTWARE THINKS: THE OPTIMIZER

3.1 The 50Hz Control Loop

Every 20 milliseconds the Raspberry Pi executes one tick of the control loop. It reads all sensors, runs 56 layered software systems in sequence, computes optimal wing commands, and sends those commands to all 8 motors and 16 servos. The 56-layer tick stack is the complete flight control system. Each layer adds one capability — stall detection, gust response, active damping, descent control — without any layer knowing about the others. The base tick produces wing commands; each wrapper refines them.

3.2 The Four-Layer Explorer

Layer 1 — Gradient	Always running. Small improvements to the current best-known configuration. Like tuning a radio.
Layer 2 — Biomimetic	Tests the 24 biological hypotheses. Each is a complete configuration derived from a real animal's locomotion.
Layer 3 — Evolutionary	Breeds new configurations from two previously good ones.
Layer 4 — Random	Explores configurations that gradient search would never reach because they require passing through worse configurations to get there.

3.3 The Demanding Input Rule

When the pilot makes a demand — any stick above 70% deflection, lever above 88% or below 12%, or abrupt control movement above 200%/s — the optimizer immediately switches to maximum performance mode. Precision, endurance, and efficiency weightings are suspended. The aircraft gives everything it has. When the pilot returns to steady inputs, normal scoring resumes after a 3-second settling period.

SECTION 4 — FULL-STATE SENSING: WHAT THE COMPUTER NOW KNOWS

4.1 The Sensing Summary

AoA (pitch)	Per wing — 8 values	MCP3208 ADC #1 on SPI1 CE1 reads each pitch servo feedback potentiometer. Actual angle updated every tick (50Hz). Enables closed-loop AoA control, per-wing stall detection, and accurate optimizer trial scoring.	FITTED
Sweep	Per wing — 8 values	MCP3208 ADC #2 on SPI1 CE0 reads each sweep servo feedback. Enables closed-loop sweep for H-16 (figure-eight stroke) and H-18 (sweep-clap). Critical for confirming the figure-eight path is executed.	FITTED
Stroke/flap	Per wing — 8 values	ODrive axis0.pos_estimate and axis1.pos_estimate give actual motor shaft position per wing. Divided by 7:1 gear ratio = actual sector wheel angle. No new hardware needed. Rope slip detected in 60ms.	NO NEW HW
Dihedral	Open-loop	Dihedral changes are slow (approach, storm stow). Aerodynamic loads on dihedral axis are low. Open-loop command is adequate.	ADEQUATE

4.2 Why This Changes Everything

Before these sensors: the optimizer was commanded to fly a configuration and assumed it was executed. If a pitch servo lagged by 4° under aerodynamic load — producing CL(14°) instead of CL(18°), a 25% difference — the optimizer would record a wrong score against the wrong configuration. Every trial had this uncertainty baked in. The gradient the optimizer computed was based on uncertain coordinates.

After: the optimizer knows what was actually flown. Trials where any axis deviated beyond tolerance are discarded before scoring. The optimizer navigates a landscape where the coordinates are known. Convergence is faster. The registry is clean. The configurations it finds are real.

SECTION 5 — SAFETY PHILOSOPHY

Rule 1: Recoverable	No automatic action and no optimizer decision may put the aircraft into a state it cannot recover from. Hard limits (3.0Hz, 96°, 22° AoA) cannot be overridden.
Rule 2: Pilot decides	Warning systems inform; the optimizer suggests; the pilot decides. The wing simulation requests approval before every test. The optimizer promotes configurations but cannot deploy them without the pilot.
Rule 3: One new thing at a time	TC-01 before TC-02. TC-02 before TC-03. TC-03 before Discovery Mode. Discovery Mode before water operations. Never add two unknowns simultaneously.

■ WARNING: The stall detection system corrects AoA on the stalling wing only. It does NOT interfere with pilot inputs above 70% stick deflection. Safety systems warn; the pilot acts.

PART II — HARDWARE

SECTION 6 — WING ASSEMBLIES

6.1 The Motor-Sector-Wheel-Rope Assembly

Each of the 8 wings has this assembly: one GoolRC 5010-14 brushless motor drives a rope that wraps around the motor output drum. The other end of the rope connects to the sector wheel. The sector wheel has a 100° mechanical arc limit. Software limits amplitude to 96° (4° safety margin). The gear ratio between motor shaft and sector wheel is set by the drum sizes — sized to achieve the target operating frequency range.

Motor model	GoolRC 5010-14, 360KV, 12N14P, 7 pole pairs
KV rating	360 rpm/V at no load
Pole pairs	7 — critical for ODrive configuration
Quantity	8 total — one per wing
Rope drive	One per wing, independent, no sharing
Sector limit	100° mechanical, 96° software limit
Vel limit	150 rad/s (set via mco_odrive_setup.py)

● CAUTION: Inspect all 8 rope drives before every flight. Fraying rope shows as increasing motor current. The stall detection Tier 2 will flag this. Rope slip is detected in 60ms from ODrive position divergence.

SECTION 7 — ODRIVE CONTROLLERS AND CAN BUS

7.1 ODrive Axis Assignment

Each ODrive v3.6 board has two independent motor outputs (axis0 and axis1) with independent velocity control, position estimation, and current sensing. Port and starboard wings of each pair connect to the two axes of one board.

Board 0	CAN node 0	axis0 = W1 port	axis1 = W1 starboard
Board 1	CAN node 1	axis0 = W2 port	axis1 = W2 starboard
Board 2	CAN node 2	axis0 = W3 port	axis1 = W3 starboard
Board 3	CAN node 3	axis0 = W4 port	axis1 = W4 starboard

7.2 What Each ODrive Axis Provides

Velocity command	odrv_nodeN.axisX.controller.input_vel — independent per wing
Position estimate	odrv_nodeN.axisX.pos_estimate — actual motor shaft position for stroke sensing
Current feedback	odrv_nodeN.axisX.motor.current_control.Iq_measured — individual wing current for stall Tier 2
Fault state	Independent per axis — W1p fault does not affect W1s or any other wing

7.3 CAN Bus Wiring

```
Pi SPI0 CE0 → MCP2515 → ODrive 0 → ODrive 1 → ODrive 2 → ODrive 3 → 120Ω
```

Daisy-chain ODrives 0→1→2→3. 120Ω termination resistor at each end of the chain. Run mco_can_setup.sh after wiring.

■ WARNING: vel_limit must be set to 150 rad/s on EVERY AXIS (8 axes total) via mco_odrive_setup.py. The factory default is lower and prevents reaching operating frequency.

SECTION 8 — PITCH SERVOS AND AoA FEEDBACK

8.1 Pitch Servo Function

Each of the 8 pitch servos rotates its wing about the spanwise axis, setting the angle of attack. AoA is the angle between the wing chord and the oncoming airflow. It determines how much lift and drag the wing generates. Having 8 independently controlled AoA values is the capability that enables H-24 (spanwise gradient), H-25 (per-stroke roll), and H-26 (force vector decoupling) — none of which have any equivalent in conventional aircraft.

Servo model	TORQ TCM-1001 or equivalent with feedback pin
Supply voltage	7.4V (2S LiPo dedicated servo rail)
PWM frequency	333Hz — high-speed digital mode required
GPIO pins	W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21
Feedback pin	REQUIRED — voltage proportional to actual position

8.2 MCP3208 ADC #1 — Pitch Feedback

The first MCP3208 reads all 8 pitch servo feedback potentiometers on SPI Bus 1, Chip Enable 1. This gives actual AoA per wing at every tick.

Chip	MCP3208-CI/P, 8-channel, 12-bit
SPI Bus	SPI1 CE1 (GPIO7 chip select)
Clock	GPIO11 (SPI1 CLK)
MOSI	GPIO10 (SPI1 MOSI)
MISO	GPIO9 (SPI1 MISO)
VREF / VDD	3.3V from Pi
Channel map	CH0=W1p, CH1=W1s, CH2=W2p, CH3=W2s, CH4=W3p, CH5=W3s, CH6=W4p, CH7=W4s
Resolution	~0.044°/step over 180° servo range
Protection	1kΩ series resistor on each signal line
Config.txt	dtoverlay=spi1-2cs

SECTION 9 — SWEEP SERVOS AND SWEEP FEEDBACK

9.1 Sweep Servo Function

Each sweep servo moves its wing fore and aft along the fuselage axis. In normal flight the sweep servo holds a fixed position. In H-16 (sea turtle figure-eight) the sweep servo moves dynamically during each stroke, tracing an elliptical path that converts some lift into thrust. In H-18 (sweep-clap) the wings converge inward in the sweep direction to produce a leading edge vortex.

Servo model	DS3225 or equivalent with feedback pin
Supply voltage	7.4V
PWM frequency	50Hz standard
GPIO pins	W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19
Feedback pin	REQUIRED for closed-loop H-16 and H-18

9.2 MCP3208 ADC #2 — Sweep Feedback

Chip	MCP3208-CI/P, identical to ADC #1
SPI Bus	SPI1 CE0 (same bus, different chip select)
Channel map	Same as pitch: CH0=W1p sweep, CH1=W1s...
Config.txt	dtoverlay=spi1-2cs (same line covers both)
Protection	1kΩ series resistor on each line

✓ NOTE: The two MCP3208 chips share SPI Bus 1 but use different chip selects (CE0 for sweep, CE1 for pitch). The single dtoverlay line enables both.

SECTION 10 — FLIGHT COMPUTER, SENSORS, AND PANEL

Raspberry Pi 4, 4GB	Flight computer. 50Hz control loop. ~11ms tick on Pi 4 (1.8× headroom vs 20ms budget).
MPU-6050 IMU	I2C, addr 0x68. Roll/pitch/yaw rates. Primary attitude.
BMP280 barometer	I2C, addr 0x76. Altitude above launch.
TF-Luna rangefinder	50Hz lidar. Height <10m. Primary for landing. Over water: reads actual wave surface.
DS18B20 OAT sensor	GPIO4, One-Wire. Outside air temperature. Density altitude correction. Enable: dtoverlay=w1-gpio,gpiopin=4.
HX711 + load cell	GPIO26 (DT), GPIO7 (SCK). TC-01/TC-02 thrust. Not flown.
Touchscreen panel	3.5" minimum. 4 buttons + 6 margin bars.
Pixhawk	Sensor bridge. GPS, compass, extra IMU. MAVLink at 921600 baud to Pi UART.

SECTION 11 — POWER SYSTEM

Main 4S LiPo	4000mAh+ for 8 motors. Powers 4 ODrives.
Servo 2S LiPo	1500mAh. Dedicated 7.4V for all 16 servos.
Pi BEC 5V 3A	Clean regulated supply — motor noise corrupts SD card.
30A main fuse	On main battery lead.
5A Pi fuse	On Pi BEC output.
Low battery	30% → amber warning. 20% → red, land now. Discovery Mode auto-exits at 20%.

■ WARNING: Use a dedicated BEC for the Pi supply. Motor PWM noise on the Pi 5V rail causes SD card write errors and can corrupt the flight log and knowledge registry.

SECTION 12 — ASSEMBLY SEQUENCE

Assemble in this order. Do not power anything until Step 11.

1	Mount airframe	Attach all structural members. Verify rigidity.
2	Mount 8 sector wheels	One per wing. Smooth rotation, no binding.
3	Mount 8 motors	One per wing. Shaft aligned with rope drum.
4	Install 8 rope drives	Thread, tension. Verify gear ratio by counting.
5	Mount 8 wings	Attach to sector wheel pivots. Full arc without binding.
6	Install 8 pitch servos	Set to neutral before attaching horn.
7	Install 8 sweep servos	Set to neutral before attaching horn.
8	Mount 4 ODrive boards	Board 0 near W1, board 3 near W4.
9	Wire motors to ODrives	W1p→Board0 axis0. W1s→Board0 axis1. W2p→Board1 axis0... etc.
10	Wire CAN bus	Board0→1→2→3 chain. 120Ω at each end. MCP2515 to Pi SPI0.
11	Wire pitch servos	Pi GPIO pitch pins. Feedback wires to MCP3208 #1.
12	Wire sweep servos	Pi GPIO sweep pins. Feedback wires to MCP3208 #2.
13	Install MCP3208 ADCs	Both chips on SPI Bus 1. CE1=pitch, CE0=sweep.
14	Mount Pi + IMU + baro	Secure Pi. Connect I2C sensors.
15	Install rangefinder	Mount facing downward, clear of wing arc.
16	Install OAT sensor	GPIO4 with 4.7kΩ pull-up to 3.3V.
17	Wire power	Main→ODrives. Servo LiPo→servo rail. BEC→Pi. Fuses.
18	Install RC receiver	SBUS to Pi UART.
19	Connect Pixhawk	TELEM2 UART to Pi at 921600 baud.
20	First power-on (Pi only)	Verify OS boots. Run mco_install.sh.
21	Configure ODrives	python3 mco_odrive_setup.py — all 8 axes.
22	Calibrate ADCs	Run servo calibration for all 16 servos.

23	Fill Measurement Sheet	Record all physical dimensions.
24	TC-01 bench test	See Section 26.

PART III — SOFTWARE INSTALLATION

SECTION 13 — RASPBERRY PI SETUP

Use Raspberry Pi Imager with Raspberry Pi OS Lite 64-bit. In advanced settings: enable SSH, set hostname to mco-1d, set WiFi credentials. After first boot:

```
sudo raspi-config
```

Enable: I2C, SPI, Serial port (disable console). Expand filesystem. Then edit /boot/config.txt:

```
# CAN adapter: dtoverlay=mcp2515-can0,oscillator=8000000,interrupt=25 # Both MCP3208 ADCs on SPI1: dtoverlay=spi1-2cs # OAT sensor: dtoverlay=w1-gpio,gpiopin=4
```

Reboot. Verify:

```
ip link show can0 # should show UP ls /dev/spidev1.* # should show spidev1.0 and spidev1.1 cat /sys/bus/w1/devices/28-*/w1_slave # OAT temperature
```

SECTION 14 — INSTALLING THE WAVE LIBRARY

```
bash mco_install.sh
```

Installs pigpio, python-can, and all dependencies. Creates /home/pi/knowledge/ for discovery registry. Installs systemd services for auto-start.

mco_wave_library.py	44,127 lines — main flight control
mco_main.py	50Hz flight loop entry point
mco_mavlink.py	Pixhawk MAVLink bridge
mco_gs_server.py	Ground station TCP server port 5760
mco_logger.py	50Hz CSV data logger
mco_odrive_setup.py	ODrive 8-axis configuration tool
mco_panel.py	Touchscreen panel driver
mco_ground_station.py	Laptop ground station client
mco_sync.py	Knowledge transfer between Pis
mco_hypothesis_tool.py	Hypothesis authoring and injection
mco_loadcell.py	HX711 reader — PENDING

SECTION 15 — CONFIGURING THE ODRIVES — ALL 8 AXES

```
python3 mco_odrive_setup.py
```

Connects to all 4 ODrive boards over CAN and configures all 8 axes. Critical settings per axis:

Motor type	MOTOR_TYPE_HIGH_CURRENT
Pole pairs	7 (GoolRC 5010-14 confirmed)
Current limit	From Build Measurement Sheet

vel_limit	150 rad/s — CRITICAL, verify saved
Control mode	CTRL_MODE_VELOCITY_CONTROL

■ WARNING: vel_limit=150 rad/s must be set on ALL 8 AXES (axis0 and axis1 on each of the 4 boards). Factory default is lower. Verify by power-cycling and re-reading the parameter.

SECTION 16 — CONFIGURING THE MCP3208 ADCs

Verify both ADCs are readable after dtoverlay is in config.txt and Pi has rebooted:

```
python3 -c "import mco_wave_library as w; print(w.read_servo_feedback_adc())"
```

Should print 8 float values (actual AoA in degrees). If [None]*8: check CE1 wiring and dtoverlay=spi1-2cs.

```
python3 -c "import mco_wave_library as w; print(w._sweep_sense_state)"
```

After connecting sweep ADC and setting sweep_adc_fitted=True, this should show actual sweep values.

SECTION 17 — GROUND STATION SETUP

```
python3 mco_ground_station.py --host mco-1d.local --port 5760
```

Displays real-time telemetry including: all 8 wing actual AoA, trim balance (port vs stbd lift), CT/CL decomposition, trial validity status, and all margin bars.

SECTION 18 — INSTALLATION VERIFICATION

V-01	Wave Library imports	python3 -c "import mco_wave_library; print(OK)" → OK
V-02	CAN bus active	ip link show can0 → UP
V-03	All 4 ODrives respond	mco_odrive_setup.py --status → 4 boards, 8 axes
V-04	IMU readable	I2C scan shows 0x68 and 0x76
V-05	Pitch ADC reads	read_servo_feedback_adc() → 8 floats (not None)
V-06	Sweep ADC reads	_sweep_sense_state shows actual values when fitted
V-07	OAT sensor	cat /sys/bus/w1/devices/28-*/w1_slave → temperature
V-08	Ground station connects	Status messages within 5 seconds
V-09	All 16 servos move	mco_odrive_setup.py --servo-test → deflection visible
V-10	8 motor axes spin	mco_odrive_setup.py --motor-test → all 8 motors

PART IV — UNDERSTANDING THE FLIGHT CONTROL SYSTEM

SECTION 19 — THE LEVER: PRIMARY FLIGHT CONTROL

The lever is the throttle stick (Channel 3). On the MCO-1D it maps to a complete wing configuration, not just power. Forward is always power, always. This convention never changes.

Above 88%	Full power/sprint zone. Demanding input rule applies. Performance only. Precision suspended.
50-88%	Normal flight. Phase-indexed scoring. Optimizer balances performance with mission objective.
~50%	Hover/neutral in flight.
12-50%	Approach/airbrake. Precision scoring highest weight.
0-12%	Steep approach. Pigeon-wing dihedral auto-activates. Glide slope 5° (at 13%) to 30° (at 0%).

SECTION 20 — THE WING CONFIGURATION SPACE

Each wing has 5 independently controllable parameters. 8 wings × 5 parameters = 40 independent dimensions. The optimizer explores this 40-dimensional space continuously during flight.

Amplitude	0-96° flapping arc per wing. Hard limit 96°.
Frequency	0.5-3.0Hz per wing. Hard limit 3.0Hz.
AoA (pitch)	0-22° per wing. S1223 CL_max at 18°. Post-stall above 22°. Now SENSED per wing.
Sweep	Fixed or dynamic (H-16). Now SENSED per wing when MCP3208 #2 fitted.
Cascade	Phase offset between pairs. Validated by ODrive pos_estimate per wing.

SECTION 21 — THE OPTIMIZER: HOW THE AIRCRAFT LEARNS

Layer 1 — Gradient	Always. Small improvements. Fast convergence.
Layer 2 — Biomimetic	All 24 hypotheses tested systematically.
Layer 3 — Evolutionary	Breeds from two good configurations.
Layer 4 — Random	Escapes local optima.

The optimizer runs continuously in all flight modes. Discovery Mode dedicates the entire flight to it. The discovery registry persists between flights on the SD card and transfers between aircraft via USB or WiFi sync.

SECTION 22 — SENSING IMPROVEMENTS TO THE OPTIMIZER

Three-axis trial validity	A trial is only scored if AoA error <3°, sweep error <5°, and stroke phase error <10°. Trials that were not accurately executed are discarded rather than corrupting the registry. Removes the largest source of discovery noise.
Trim balance objective	Actual port vs starboard lift asymmetry computed every tick. The optimizer rewards configurations that produce zero roll trim offset. A balanced aircraft has full differential amplitude authority available for roll and gust response.
CT/CL decomposition	Actual thrust (horizontal) and lift (vertical) computed from per-wing lift estimates and actual AoA. Optimizer can target the thrust-lift ratio needed for each phase: hover (CL), transit (CT+CL), paddle (CT).
Servo bandwidth constraint	After H-27 data is collected: exploration constrained to AoA rate-of-change achievable within measured servo speed. Eliminates physically impossible configurations from search.
Symmetry fault detection	If a symmetric configuration shows port/stbd AoA difference >2°: flagged as servo issue, not configuration property. Prevents one slow servo from penalising a good configuration.

SECTION 23 — THE BIOMIMETIC HYPOTHESIS LIBRARY

H-01	Dragonfly	Phase switching — in-phase for force, 5v
H-02	Bat advance	Pre-stroke AoA transition. Direct Re match. 5v
H-03	Manta wave	Spanwise phase gradient root-to-tip. 3v
H-04	Pterosaur soar	0.8Hz, 18° AoA (Re-corrected from 32°). 1v
H-05	Hawkmoth LEV	Amplitude gradient delays LEV breakdown. 4v
H-06	Neg. cascade	W4 leads W1 — reversed cascade. 1v
H-07	Split freq.	10% inter-pair frequency differential. 3v
H-08	Strouhal	Speed-adaptive St 0.20-0.40. speed_dependent
H-12	Locust burst	ODrive velocity burst at reversal. 3v
H-13	Plesiosaur phase	Speed-adaptive cascade phase. 4v
H-14	Plesiosaur brake	Anti-augmentation drag. 1v
H-15	Plesiosaur dom.	Pair amplitude dominance switching. 6v
H-16	Turtle figure-8	Dynamic sweep per stroke. Needs sweep ADC. 4v
H-17	Turtle asymm.	Power down, recovery up. 5° floor (air corrected). 1v
H-18	Turtle clap	Sweep-axis clap at downstroke bottom. 3v
H-19	Sea lion AoA	15° thrust, 22° lift (Re-corrected). 4v
H-20	Otter A-ratio	Drag-to-lift mode switch with speed. 5v
H-21	Sea lion fling	High-velocity clap initiation. 3v
H-22	Bat upstroke	Ventral clap double lift peak. Direct Re. 4v
H-23	Bat AoA adapt.	-8° hover → +5° cruise → -3° sprint. Direct Re. 6v
H-24	AoA gradient	MCO-1D ORIGINAL. Spanwise AoA gradient. 4v
H-25	Per-stroke roll	MCO-1D ORIGINAL. AoA-based roll, 4x faster. 4v
H-26	Force decouple	MCO-1D ORIGINAL. Sidestep without bank. 4v

H-27	Servo BW map	MCO-1D ORIGINAL. Needs MCP3208. 12v

Total: 24 active hypotheses, 90 variants. H-24 to H-27 are MCO-1D originals with no biological precedent. H-16, H-18, H-25, H-26, H-27 require or benefit from the per-wing sensing suite.

SECTION 24 — FAULT TOLERANCE

255 pre-computed combinations	Every possible combination of failed wings (1 to 7) is pre-computed at startup. Response is immediate.
Per-wing isolation	A port wing failure does not affect its starboard partner. With 8 independent motors, each failure affects only one wing.
Urgency language	CONTINUE WITH CAUTION (7 wings healthy). LAND AS SOON AS PRACTICABLE (6 wings). LAND AS SOON AS POSSIBLE (5 wings or worse). LAND IMMEDIATELY (uncontrollable configuration).
Rope slip detection	ODrive pos_estimate divergence >5° from model detected in 60ms. Pilot notification before hard failure.
Servo health trending	AoA tracking RMS per wing. Rising error = servo degrading. Warning before hard failure.

```
failure_briefing # pre-flight failure performance table fault_summary # current fault and degradation state servo_health_report # AoA tracking RMS all 8 wings
```

SECTION 25 — THE MARGIN TRACKER

Speed margin	Current airspeed vs demonstrated Vmax
Stall margin	Above stall speed, corrected for bank angle
Frequency margin	Current freq vs 3.0Hz ceiling
Amplitude margin	Current amp vs 96° limit
Gust margin	Turbulence vs survivable conditions + amplitude headroom
Combined margin	Minimum of all five — the coffin corner metric

■ WARNING: The margin tracker warns only. It does not take automatic action. Exploration pauses when margins are narrow. All control remains with the pilot at all times.

PART V — COMMISSIONING TESTS

SECTION 26 — TC-01: BENCH TEST

TC-01 is a ground test. The aircraft does not leave the bench. Purpose: verify every piece of hardware works before applying aerodynamic loads. All TC-02 and TC-03 assumptions depend on TC-01 passing completely.

26.1 TC-01 Test Sequence

TC-01-01	Servo neutral check	All 16 servos at commanded neutral. Measure actual angle. Record deviation. Adjust trim offsets if >2°.
TC-01-02	Servo range check	Command each servo to ±45°. Verify travel, no binding.
TC-01-03	Pitch ADC verification	python3 -c "import mco_wave_library as w; print(w.read_servo_feedback_adc())" — must show 8 floats.
TC-01-04	Sweep ADC verification	Same as TC-01-03 for sweep ADC channel.
TC-01-05	Pitch servo feedback calibration	For each pitch servo: command to -45°, 0°, +45°. Record ADC voltage at each. Call calibrate_servo_feedback(). Repeat for all 8 wings (indices 0-7).
TC-01-06	Sweep servo feedback calibration	Same procedure for all 8 sweep servos.
TC-01-07	ODrive 8-axis communication	mco_odrive_setup.py --status — all 4 boards, all 8 axes respond.
TC-01-08	Motor spin test — low speed	Command each of 8 axes to 10 rad/s. Listen for bearing noise.
TC-01-09	Wing flapping at 0.5Hz	All 8 wings at 0.5Hz. Verify smooth mechanical motion. Check all 8 rope drives for slip or fraying.
TC-01-10	Wing flapping at operating frequency	All 8 wings at 2.5Hz (estimated hover). Run 60 seconds. Monitor motor temperatures. Verify ADC reads valid AoA.
TC-01-11	Frequency ceiling test	Step 2.0→3.0Hz in 0.1Hz steps. Hold 30s each. Record motor temperatures. Identify thermal limit. This is the operational frequency ceiling.

TC-01-12	Static thrust measurement	Secure to load cell. Command 2.5Hz estimated hover. Record thrust. Vary 1.5-3.0Hz, record at each step. Must exceed MTOW (33.8N) at some frequency.
TC-01-13	RC channel mapping	Move all sticks to extremes. Verify CH1=roll, CH2=pitch, CH3=lever, CH4=yaw.
TC-01-14	Safety system verification	Command freq >3.0Hz → verify hard cap activates. Command amp >96° → verify hard cap activates. Unplug one motor → verify fault detection fires.
TC-01-15	Symmetry verification	Command all wings to identical AoA (15°). Run lift_balance_report. Port and stbd within 5%.
TC-01-16	Fill Measurement Sheet	Record: hover thrust, frequency ceiling, motor temps, servo calibration voltages, rope drive condition.

26.2 TC-01 Pass Criteria

All 16 servos	Within ±2° of commanded, both axes (pitch and sweep)
Both ADCs	8 valid readings each, calibrated
All 8 motors	Spin without unusual noise, no fault codes
Frequency ceiling	Sustained 2.5Hz minimum for 60 seconds
Static thrust	Exceeds 33.8N (MTOW) at some frequency
Safety caps	Both hard caps confirmed active
Symmetry	Port/stbd lift within 5% at same AoA

■ WARNING: Do not proceed to TC-02 unless ALL TC-01 criteria pass. If static thrust does not exceed MTOW at the frequency ceiling, the aircraft cannot hover.

SECTION 27 — TC-02: HOVER TEST

TC-02 is the first flight. Goal: controlled stable hover for a measured time. Every advanced system depends on data only TC-02 can produce. Two people minimum: pilot on transmitter, operator at ground station laptop.

27.1 Site Requirements

Area	50m × 50m clear minimum. No people within 30m.
Wind	Below 5 knots. No gusts. TC-02 in calm only.
Surface	Grass or soft ground preferred.
Elevation	Sea level to 300m. Not higher for first flight.

27.2 TC-02 Test Sequence

TC-02-01	First hover	Increase lever slowly 50%→70%. Record lever and freq at lift-off. This is the hover lever position. Land. Repeat 3 times.
TC-02-02	Hover stability	Hold hover 30 seconds without correction. sigma_h target <0.5m. If exceeded: investigate before continuing.
TC-02-03	Measure hover frequency	Hover 60 seconds. Read actual freq_hz from ground station telemetry. Record: this is DESCENT_CTRL_HOVER_FREQ_HZ.
TC-02-04	Lift-frequency curve	At constant altitude: vary freq ±0.3Hz from hover. Record altitude change rate per step. Record: lift_freq_slope_N_per_Hz for calibrate_descent_controller().
TC-02-05	Roll response test	Apply 20%, 40%, 60% roll stick from hover. Record roll rate. Release. Record settle time. Record: calibrates ACTIVE_DAMP_K_AMP and SETTLE_TIME_S.
TC-02-06	Calibrate descent controller	calibrate_descent_controller(hover_freq_hz=X, lift_freq_slope_N_per_Hz=Y) Verify 100 fpm descent is achievable and stable.

TC-02-07	H-26 sidestep observation	From hover: command small differential AoA port/stbd (H-26). Observe: does aircraft translate laterally or roll? This is the key unknown — unknown until this moment.
TC-02-08	First discovery	Enable auto-discovery. Hover 5 minutes. Allow optimizer to begin building hover configuration map.

27.3 Critical TC-02 Calibration Outputs

Hover frequency	DESCENT_CTRL_HOVER_FREQ_HZ — scales all descent gains
Lift-freq slope	dL/df at hover — for calibrate_descent_controller()
Roll rate	deg/s per % stick — scales ACTIVE_DAMP_K_AMP
Settle time	seconds to <2°/s — scales ACTIVE_DAMP_SETTLE_TIME_S
H-26 result	translate or roll? Unknown until this flight.
AoA CL gains	AOA_CL_KP, KI, KD for closed-loop AoA controller

SECTION 28 — TC-03: TRANSITION AND FORWARD FLIGHT

TC-03-01	Transition	From hover, advance lever toward 70%. Allow pitch to develop. Record transition speed and lever position.
TC-03-02	Forward flight altitude hold	Hold altitude using descent controller at cruise lever. Record airspeed, power consumption.
TC-03-03	Speed envelope	Fly at 60-80% lever steps. Record airspeed and power. Identify minimum stable speed (stall speed in forward flight).
TC-03-04	IFR descent validation	Engage descent_ctrl_engage 100 in forward flight. Verify 100 fpm maintained. Verify 300 fpm when commanded.
TC-03-05	First Discovery Mode flight	Fly racetrack at 100ft AGL. Engage disc_mode_engage. Run full 60-minute session. Export knowledge.
TC-03-06	CT-J curve	Measure thrust coefficient vs advance ratio across speed range. Required for stroke plane controller design (post-TC-03).

PART VI — FLIGHT OPERATIONS

SECTION 29 — NORMAL PRE-FLIGHT PROCEDURE

1	Area	Clear of people. Landing zone identified.
2	Weather	Below 15kts normal ops. Below 5kts for TC-02.
3	Battery	Main >90%. Servo >90%. Check on ground station.
4	ODrives	Ground station shows all 8 axes IDLE.
5	ADC check	servo_health_report shows all 8 wings healthy.
6	RC signal	Ground station shows RC VALID.
7	IMU	Aircraft level. Reads 0° pitch, 0° roll.
8	Knowledge	Discovery registry status shown on ground station.
9	Margins	All 6 margin bars green.
10	Arm	Press ARM. Verify armed state.
11	Control check	All sticks: verify correct response direction.

SECTION 30 — NORMAL FLIGHT

Takeoff	Advance lever smoothly from 50% toward hover lever position found in TC-02. Do not jerk — smooth input gives the optimizer time to find the right configuration.
Hover	Hold lever at hover position. Gentle roll/yaw for positioning. The trim balance system ensures port and stbd contribute equally.
Forward flight	Advance lever from hover toward 65-70%. Allow pitch to develop. Hold altitude with lever position.
IFR approach	descent_ctrl_engage 100. Target 100 fpm. Rangefinder provides height reference. Do not flatten out — continue descent until touchdown.
Landing	Reduce lever toward 50%. Hold off. Reduce lever to land. Do not cut lever abruptly — aircraft needs a few wingbeats to lose momentum cleanly.

SECTION 31 — EMERGENCY PROCEDURES

31.1 Loss of Control

■ WARNING: If aircraft moves unexpectedly: reduce lever toward 50%. If continues: full lever aft. Disarm if not controllable. A hard landing is better than a flight endangering people.

31.2 Wing Failure

Read urgency level from ground station. The fault system has already applied compensation — continue to the nearest safe landing point at the urgency level indicated. With 8 independent motors, single-wing faults are well within the compensation envelope at 7 remaining wings.

31.3 Battery Low

30% — amber	Land soon. Complete current objective.
20% — red	Land now. Abandon objective. Discovery Mode exits.

SECTION 32 — POST-FLIGHT

1	Disarm	Press DISARM. Motors stop.
2	Cooldown	Press POST-FLIGHT. 8-minute motor cooling.
3	Export	knowledge_export_usb or sync to laptop.
4	Logs	Copy /home/pi/logs/ to backup.
5	Battery	Check for puffing or unusual heat.

6	Rope drives	Inspect all 8 for fraying or slipping.
7	ADC check	servo_health_report — verify no trending degradation.
8	Airframe	Visual inspection for cracks.

PART VII — DISCOVERY PROGRAMME

SECTION 33 — DISCOVERY MODE

Discovery Mode dedicates the entire flight to exploration. The aircraft holds a racetrack pattern at 100 feet and runs all four exploration layers at maximum throughput. The three-axis trial validity gate ensures only accurately executed configurations are scored.

```
disc_mode_engage(inbound_heading=270, duration_min=60) disc_mode_engage_focused("biomimetic")
# hypothesis focus disc_mode_engage_focused("precision") # sigma minimisation
disc_mode_engage_focused("aerobatic") # crispness and stability disc_mode_exit # exit and
export
```

Focus modes available:

full	All four blocks in rotation (default)
gust_resilience	Turbulence stability in real wind
fault_tolerance	Wing simulation and progressive chains
cruise_efficiency	Range per watt-hour at transit speeds
hover_endurance	Minimum power for SAR loiter
precision	Minimise sigma h, roll, hdg, sink
aerobatic	Crispness and stability
water_operations	Wave timing and approach
biomimetic	All 24 hypotheses at full rate
envelope	Envelope expansion only

SECTION 34 — WING SIMULATION AND PROGRESSIVE FAILURE

```
wing_sim_request # show next untested combination wing_sim_approve # arm test for next
suitable conditions prog_sim_request # next priority progressive chain prog_sim_approve # arm
chain test at 250ft minimum
```

255 failure combinations pre-computed. 2,080 progressive chains up to 4 steps long. With 8 independent motors, each failure is truly isolated — wing sim tests are more representative than with shared drive systems.

SECTION 35 — PRECISION AEROBATICS

Active damping detects stick-to-neutral with residual roll rate and immediately applies opposing differential amplitude. The aircraft snaps to the current attitude and holds it. With actual AoA feedback the damping force is computed from actual wing state, preventing overshoot.

```
aerobatic_status # crispness score and hold stability
```

Eight-point hesitation roll: push stick → roll 45° → release → lock → push → roll 45° → release → lock. Repeat 8 times. The pilot does the timing; the system does the holding.

SECTION 36 — WATER OPERATIONS

● CAUTION: Water operations require float hardware not yet specified. Hover must be demonstrated at TC-02 first. The software framework is complete and ready.

Wave limits	15 inches (0.38m) FAA safe. 18 inches (0.46m) restriction.
Departure	Apply power as float RISES toward crest (FAA H-8083-23). Never from trough.
Glassy water	descent_ctrl_engage 100. Rangefinder only. No visual reference. Continue to touchdown.
Porpoising	First osc: maintain back pressure. Second osc: power to idle, full elevator back.

PART VIII — REFERENCE

SECTION 37 — GROUND STATION COMMANDS (SELECTED)

arm / disarm	Arm or disarm all 8 motors
emergency_stop	Immediate wing stop
failure_briefing	Pre-flight failure performance table
fault_summary	Current fault and degradation state
servo_health_report	AoA tracking RMS all 8 wings
lift_balance_report	Port vs stbd actual lift per wing
stroke_position_status	Actual vs commanded stroke all 8 wings
stall_detector_status	Per-wing stall likelihood
trim_balance status	Port/stbd balance and asymmetry
disc_mode_engage	Engage Discovery Mode
disc_mode_status	Discovery session status
descent_ctrl_engage N	Engage descent at N fpm (100 default)
descent_ctrl_status	Descent controller state
aerobatic_status	Active damping and aerobatic score
precision_status	Sigma values and precision score
wave_status	Wave height and period estimate
knowledge_export_usb	Export discovery registry to USB
calibrate_servo_feedback	Per-servo ADC voltage calibration
update_servo_bandwidth	Store H-27 bandwidth measurement

SECTION 38 — GPIO AND ADC PIN REFERENCE

GPIO 4	DS18B20 OAT (One-Wire)
GPIO 5	W3p sweep servo PWM
GPIO 6	W3s sweep servo PWM
GPIO 7	MCP3208 #1 CE1 (pitch ADC chip select)
GPIO 8	MCP3208 #2 CE0 (sweep ADC chip select)
GPIO 9	SPI1 MISO (shared by both ADCs)
GPIO 10	SPI1 MOSI (shared by both ADCs)
GPIO 11	SPI1 CLK (shared by both ADCs)
GPIO 12	W3p pitch servo PWM
GPIO 13	W4p sweep servo PWM
GPIO 16	W3s pitch servo PWM
GPIO 17	W1s sweep servo PWM
GPIO 18	W1p pitch servo PWM
GPIO 19	W4s sweep servo PWM
GPIO 20	W4p pitch servo PWM
GPIO 21	W4s pitch servo PWM
GPIO 22	W2s sweep servo PWM
GPIO 23	W1s pitch servo PWM
GPIO 24	W2p pitch servo PWM
GPIO 25	W2s pitch servo PWM / MCP2515 CAN interrupt
GPIO 26	HX711 load cell DT
GPIO 27	W2p sweep servo PWM
SPI0	MCP2515 CAN adapter (ODrive CAN bus)
SPI1 CE0	MCP3208 #2 sweep servo feedback ADC
SPI1 CE1	MCP3208 #1 pitch servo feedback ADC
I2C1	IMU (0x68) and barometer (0x76)
UART0	Pixhawk MAVLink at 921600 baud

SECTION 39 — ODRIVE AXIS ASSIGNMENT

Board 0	Node 0	axis0	W1 port (W1p)	axis1	W1 starboard (W1s)
Board 1	Node 1	axis0	W2 port (W2p)	axis1	W2 starboard (W2s)
Board 2	Node 2	axis0	W3 port (W3p)	axis1	W3 starboard (W3s)
Board 3	Node 3	axis0	W4 port (W4p)	axis1	W4 starboard (W4s)

Per-axis CAN messages (velocity control):

```
# Send velocity command to W2 starboard (Board 1, axis1): node1.axis1.controller.input_vel = target_rad_s # Read W2 starboard stroke position: pos = node1.axis1.pos_estimate # motor shaft radians sector_deg = pos * 180/pi / 7 # sector wheel degrees
```

SECTION 40 — SOFTWARE FILE REFERENCE

mco_wave_library.py	44,127 lines. Main flight control. 56-layer tick stack. All systems.
mco_main.py	~307 lines. 50Hz flight loop.
mco_mavlink.py	~290 lines. Pixhawk bridge.
mco_gs_server.py	~193 lines. TCP server port 5760.
mco_logger.py	~185 lines. 50Hz CSV logger.
mco_odrive_setup.py	~187 lines. 8-axis ODrive config.
mco_panel.py	~485 lines. Panel driver.
mco_ground_station.py	~588 lines. Laptop client.
mco_sync.py	~229 lines. Knowledge transfer.
mco_hypothesis_tool.py	~287 lines. Hypothesis authoring.
mco_loadcell.py	PENDING — HX711 for TC-01/TC-02.
mco_can_setup.sh	CAN interface setup script.
mco_install.sh	One-step installation.

SECTION 41 — TC-02 CALIBRATION ITEMS

All items below are UNKNOWN until TC-02 first flight. Current values are physics-based estimates.

DESCENT_CTRL_HOVER_FREQ_HZ	2.50 Hz est.	Measure: hover 60s, read freq_hz from telemetry. Range expected: 1.8-3.2Hz.
DESCENT_CTRL_K_FREQ_PER_FPM	0.0009 est.	Derive: from hover freq and lift-freq slope via calibrate_descent_controller().
ACTIVE_DAMP_K_AMP	0.40 est.	Measure: roll rate per % stick. Calibrate from roll response test TC-02-05.
ACTIVE_DAMP_K_AOA	0.15 est.	Calibrate together with K_AMP from TC-02-05.
ACTIVE_DAMP_SETTLE_TIME_S	0.15s est.	Measure: time from stick neutral to rate <2°/s. TC-02-05 roll response test.
AOA_CL_KP / KI / KD	0.30/0.05/0.02 est.	AoA closed-loop PID gains. Tune from step response with MCP3208 feedback active.
H-26 behaviour	Unknown	Does differential AoA produce lateral translation or roll? Only TC-02 can answer this.

SECTION 42 — STATISTICS AND OUTSTANDING ITEMS

Wave Library	44,127 lines Python
Tick stack depth	56 layered wrappers
Tick time (dev)	0.90ms mean, 1.37ms P99
Tick time (Pi 4 est)	~11ms mean, 1.8× headroom vs 20ms
CL/CD cache hit rate	96%+ after warm-up
Motor count	8 (corrected from previous 4)
Active hypotheses	24 (H-01 to H-27, excl H-09/10/11)
H-09/10/11 status	Pending ODrive PWM validation (H-09), bench rope test (H-10), servo speed (H-11)
Hypothesis variants	90 total
Failure combinations	255 pre-computed
Progressive chains	2,080 generated
GS commands	105 registered
Per-wing sensing	AoA ✓ Sweep ✓ Stroke ✓
mco_loadcell.py	PENDING before TC-01
Float specification	PENDING after TC-02 hover confirmation
Stroke plane ctrllr	PENDING after TC-03 CT-J curve
GPS position hold	PENDING for racetrack precision